

Association of polychlorinated biphenyls with hypertension in the 1999-2002 National Health and Nutrition Examination Survey.



The association of 11 polychlorinated biphenyls (PCBs) with hypertension was investigated using the National Health and Nutrition Examination Survey (NHANES), 1999-2002. The unweighted number of participants assessed for hypertension ranged from 2074 to 2556 depending on the chemical(s) being analyzed. In unadjusted logistic regressions all 11 PCBs were associated with hypertension. After adjustment for age, gender, race, smoking status, body mass index, exercise, total cholesterol, and family history of coronary heart disease, seven of the 11 PCBs (PCBs 126, 74, 118, 99, 138/158, 170, and 187) were significantly associated with hypertension. The strongest adjusted associations with hypertension were found for dioxin-like PCBs 126 and 118. PCB 126 >59.1 pg/g lipid adjusted had an odds ratio of 2.45 (95% CI 1.48-4.04) compared to PCB 126 ≤26.1 pg/g lipid adjusted. PCB 118 >27.5 ng/g lipid adjusted had an odds ratio of 2.30 (95% CI 1.29-4.08) compared to PCB 118 ≤12.5 ng/g lipid adjusted. Moreover, participants with one or more elevated PCBs had an odds ratio of 1.84 (95% CI 1.25-2.70) compared to no PCBs elevated in an adjusted logistic regression. The prevalence of one or more elevated PCBs was 22.76% or 32 million of 142 million persons ≥20 years old in the non-institutionalized US population. We hypothesize that association of seven PCBs with hypertension indicates elevated PCBs are a risk factor for hypertension. What clinicians can do, given the results of this study, is limited unless the appropriate laboratory methods can be made more widely available for testing patients.